

# Solar N appendix

Sun · N · downloadable appendix

Solar nitrogen:  $7.866 \pm 0.194$

Asplund, Amarsi & Grevesse 2021:  $7.83 \pm 0.07$  · this measurement +0.036 dex

The solar N I lines are weak (0.3–1% deep) and carry CN absorption inside their own profiles, so the result depends on molecular opacity and on continuum placement at the 0.1% level. Both are now modelled: CN is synthesised with the line, and the continuum is placed locally around every line. For comparison, Asplund et al. 2021 give 7.77 from N I after empirically rescaling the CN blends, and Lodders, Bergemann & Palme 2025 adopt  $7.94 \pm 0.11$  from the two least-blended lines. The error bar carries the full uncertainty budget, including CN blending and telluric residuals.

Kitt Peak · NIR · Reference Grade · MOL-CN\_AX\_IR · 1D-LTE · n = 215. The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

## Highlighted Products

### red-optical

**$7.907 \pm 0.244$**

$\pm 0.144$  stat ·  $\pm 0.042$  syst

SET-AGSS21 · N I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied · solar\_kpno\_molecfits\_corrected · Reference Grade

### NIR

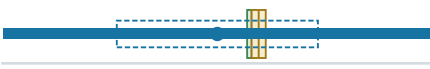
**$7.866 \pm 0.194$**

$\pm 0.024$  stat ·  $\pm 0.193$  syst

MOL-CN\_AX\_IR · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied · solar\_kpno\_molecfits\_corrected · Reference Grade

## Error-bar Forest Plot

### VIS — harps solar\_harps\_molecfits\_corrected A(N) dex; axis 3.93–10.66

|                                                                                                         |                                                                                      |                                            |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------|
| MOL-CN_red · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied<br>Codex Grade · ALL · n=194 |  | 7.296<br>$\pm 3.338$ stat $\pm 1.567$ syst |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------|

VIS axis: 3.93 · 5.61 · 7.30 · 8.98 · 10.66 A(N) dex

### red-optical — iag\_fts\_solar\_atlas solar\_iag A(N) dex; axis 7.73–8.38

|                                                                                                                             |                                                                                      |                                            |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------|
| SET-LBP25 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied<br>Reference Grade · ALL · n=2                    |  | 7.949<br>$\pm 0.010$ stat $\pm 0.042$ syst |
| SET-LBP25 · N I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied<br>Reference Grade · ALL · n=2 |  | 7.904<br>$\pm 0.010$ stat $\pm 0.042$ syst |
| SET-LBP25 · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied<br>Reference Grade · ALL · n=1     |  | 7.935<br>$\pm 0.010$ stat $\pm 0.042$ syst |

## red-optical — kpno\_solar\_atlas solar\_kpno\_kurucz2005\_corrected A(N) dex; axis 7.73-8.38

SET-LBP25 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE ·  
line set not supplied  
Reference Grade · ALL · n=2



7.939  
±0.013 stat ±0.042 syst

SET-LBP25 · N I · Synth · 3D-NLTE · Amarsi · SYNTH ·  
ENGINE-A-3DNLTE · line set not supplied  
Reference Grade · ALL · n=2



7.894  
±0.010 stat ±0.042 syst

SET-LBP25 · N I · Synth · 1D-NLTE · Bergemann ·  
SYNTH · ENGINE-A · line set not supplied  
Reference Grade · ALL · n=1



7.912  
±0.010 stat ±0.042 syst

## red-optical — kpno\_solar\_atlas solar\_kpno\_molecfit\_corrected A(N) dex; axis 7.73-8.38

SET-AGSS21 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE ·  
line set not supplied  
Reference Grade · ALL · n=3



7.960  
±0.142 stat ±0.042 syst

SET-AGSS21 · N I · Synth · 3D-NLTE · Amarsi · SYNTH ·  
ENGINE-A-3DNLTE · line set not supplied  
Reference Grade · ALL · n=3



7.907  
±0.144 stat ±0.042 syst

SET-AGSS21 · N I · Synth · 1D-NLTE · Bergemann ·  
SYNTH · ENGINE-A · line set not supplied  
Reference Grade · ALL · n=2



8.132  
±0.222 stat ±0.042 syst

SET-LBP25 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE ·  
line set not supplied  
Reference Grade · ALL · n=2



7.942  
±0.018 stat ±0.042 syst

SET-LBP25 · N I · Synth · 3D-NLTE · Amarsi · SYNTH ·  
ENGINE-A-3DNLTE · line set not supplied  
Reference Grade · ALL · n=2



7.897  
±0.010 stat ±0.042 syst

SET-LBP25 · N I · Synth · 1D-NLTE · Bergemann ·  
SYNTH · ENGINE-A · line set not supplied  
Reference Grade · ALL · n=1



7.910  
±0.010 stat ±0.042 syst

red-optical axis: 7.73 · 7.89 · 8.06 · 8.22 · 8.38 A(N) dex

## NIR — crires\_plus solar\_crires\_plus\_j\_rya1219 A(N) dex; axis 7.64-8.25

MOL-CN\_AX\_IR · N I · Synth · 1D-LTE · SYNTH ·  
1D-LTE · line set not supplied  
Reference Grade · ALL · n=185



8.035  
±0.078 stat ±0.181 syst

## NIR — kpno\_solar\_atlas solar\_kpno\_molecfit\_corrected A(N) dex; axis 7.64-8.25

MOL-CN\_AX\_IR · N I · Synth · 1D-LTE · SYNTH ·  
1D-LTE · line set not supplied  
Reference Grade · ALL · n=215



7.866  
±0.024 stat ±0.193 syst

NIR axis: 7.64 · 7.79 · 7.94 · 8.10 · 8.25 A(N) dex

**How to read this forest.** Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

## How this was measured

Every nitrogen value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the nitrogen abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, C<sub>2</sub>, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

## Solar nitrogen context

The nitrogen strategy uses red-optical N I as the primary atomic route, with NH providing an independent molecular check and CN a carbon-dependent cross-check. An unresolved or unconstrained fit supplies no adopted Solar nitrogen abundance.

[C/O indicator strategy](#) · [Methodology](#) · [Nitrogen strategy](#)

## Open provenance decisions

NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation.

[Adjudication record](#) · [Indicator gf checks](#)

## Download the evidence

[Download PDF — Solar N appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

## Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit

56719920b830f9778d50a22bd7a9604a6ddc7c28

Feed N.json v1.79 · updated 2026-10-09T08:09:24Z

Generated 2026-10-10T09:46:17-06:00 (source commit timestamp)

Feed SHA-256 3561f370e12e0324384eb4931f44b043dc5e49a4696ab7ed58a17afca923faeb

Generator scripts/generate\_cno\_publication.py v1.0.0

[Pinned scientific source](#)

## References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

**Solar abundance and indicator analyses.** Solar N atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

**Atomic data and gf authorities.** The recorded N I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [3], [4].

**Atomic data and gf authorities.** NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation. Sources: [3], [5].

**Model and grid sources.** The N I registry uses the Amarsi GALAH/PySME 1D-NLTE departure grid. The solar 3D-NLTE atomic analysis is a separate source, not a full-3D correction applied by this feed. Sources: [6], [7], [8].

**Molecular line data and delivery.** CN line-data delivery: CN red A-X: Brooke et al. 2014 / Sneden et al. 2014 (isotopologues). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpd.lmpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [9], [10], [11].

**Model and grid sources.** Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [9].

**Atomic data and gf authorities.** Atomic N I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4], [5].

**Model and grid sources.** Feed label: Synth · 3D-NLTE · Amarsi. Cited per-line solar correction, Amarsi et al. 2020 Table 3 (3D non-LTE minus 1D LTE); not an in-transfer 3D-NLTE result. Run provenance: Amarsi, Grevesse, Grumer, Asplund, Barklem & Collet 2020, A&A 636, A120, Table 3 (solar, disk-centre EWs) -- 3D non-LTE (STAGGER) - 1D LTE (MARCS); SOLAR per-line additive correction, same EWs in both Sources: [6], [7], [8].

**Model and grid sources.** Feed label: Synth · 1D-NLTE · Bergemann. Registered source: Amarsi/PySME N I 1D-NLTE. The generic Bergemann label is retained as supplied and is not used as a bibliographic attribution. Sources: [6], [7], [8].

Product-to-source index

N-products-f80b21113bb7 · products · NIR · solar\_crires\_plus\_j\_rya1219 · MOL-CN\_AX\_IR · 1D-LTE · not supplied · [1], [2], [9], [10], [11]

N-products-bbb7416bb60c · products · NIR · solar\_kpno\_molecfits\_corrected · MOL-CN\_AX\_IR · 1D-LTE · not supplied · [1], [2], [9], [10], [11]

N-products-04b24d6c2d67 · products · VIS · solar\_harps\_molecfits\_corrected · MOL-CN\_red · 1D-LTE · not supplied · [1], [2], [9], [10], [11]

N-products-854e3ddc73f8 · products · red-optical · solar\_iag · SET-LBP25 · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

N-products-95a9d8d61cb1 · products · red-optical · solar\_iag · SET-LBP25 · ENGINE-A-3DNLTE · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-74b4db6b5e6a · products · red-optical · solar\_iag · SET-LBP25 · ENGINE-A · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-9ad55347df16 · products · red-optical · solar\_kpno\_kurucz2005\_corrected · SET-LBP25 · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

N-products-3feb158ef160 · products · red-optical · solar\_kpno\_kurucz2005\_corrected · SET-LBP25 · ENGINE-A-3DNLTE · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-c22e5b580ebb · products · red-optical · solar\_kpno\_kurucz2005\_corrected · SET-LBP25 · ENGINE-A · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-3034d4211c0b · products · red-optical · solar\_kpno\_molecfits\_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

N-products-f201c06612d2 · products · red-optical · solar\_kpno\_molecfits\_corrected · SET-AGSS21 · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-products-71ab16a32bba · products · red-optical · solar\_kpno\_molecfits\_corrected · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-products-b5ebbb0c7d12 · products · red-optical · solar\_kpno\_molecfits\_corrected · SET-LBP25 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

N-products-f53ab893b6e5 · products · red-optical · solar\_kpno\_molecfits\_corrected · SET-LBP25 · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-products-5bf5dcb5e0cd · products · red-optical · solar\_kpno\_molecfits\_corrected · SET-LBP25 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-quarantine-fc1525ccd677 · quarantine · red-optical · solar\_iag · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

N-quarantine-9cb72bc628d8 · quarantine · red-optical · solar\_kpno\_kurucz2005\_corrected · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)

Complete applicable bibliography

## Solar abundance and indicator analyses

**[1]** — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. *A&A* 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

**[2]** — Amarsi et al. 2021, *A&A* 656 A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

## Atomic data and gf authorities

**[3]** — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. [source record](#)

**[4]** — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) [source record](#)

**[5]** — Tachiev & Froese Fischer 2002, *A&A* 385, 716 DOI: 10.1051/0004-6361:20011816. [source record](#)

## Model and grid sources

**[6]** — Amarsi et al. 2020, *A&A* 642 A62 DOI: 10.1051/0004-6361/202038650. [source record](#)

**[7]** — Amarsi, Grevesse, Gruber, Asplund, Barklem & Collet 2020, *A&A* 636, A120 DOI: 10.1051/0004-6361/202037890. [source record](#)

**[8]** — Amarsi 'Grid/NLTE' Zenodo, version 3 (the original 13-element 2020 release) DOI: 10.5281/zenodo.3982506. [source record](#)

## Molecular line data and delivery

**[9]** — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. *A&A* 669, A43 DOI: 10.1051/0004-6361/202243673. [source record](#)

**[10]** — Brooke et al. 2014, *ApJS* 210, 23 DOI: 10.1088/0067-0049/210/2/23. [source record](#)

**[11]** — Sneden, C.; Lucatello, S.; Ram, R. S.; Brooke, J. S. A.; Bernath, P. (2014). Line Lists for the Red and Violet Systems of CN, <sup>13</sup>C<sup>14</sup>N, and <sup>12</sup>C<sup>15</sup>N, and Application to Astronomical Spectra. *ApJS* 214, 26. DOI: 10.1088/0067-0049/214/2/26. [source record](#)